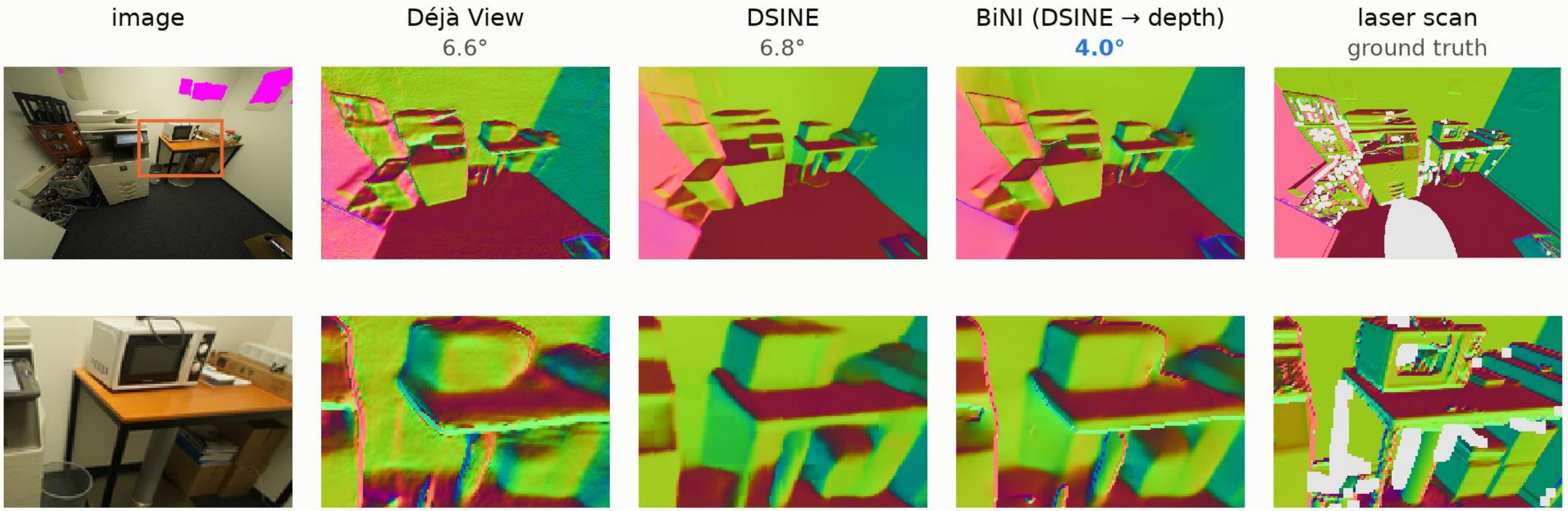
DSINE + BiNI: sharper normals, not more accurate depth

BiNI integrates DSINE's per-pixel normals into depth, held near Déjà View's; no other optimization.



zoom (orange box) · ° = median angle to the scan's normals

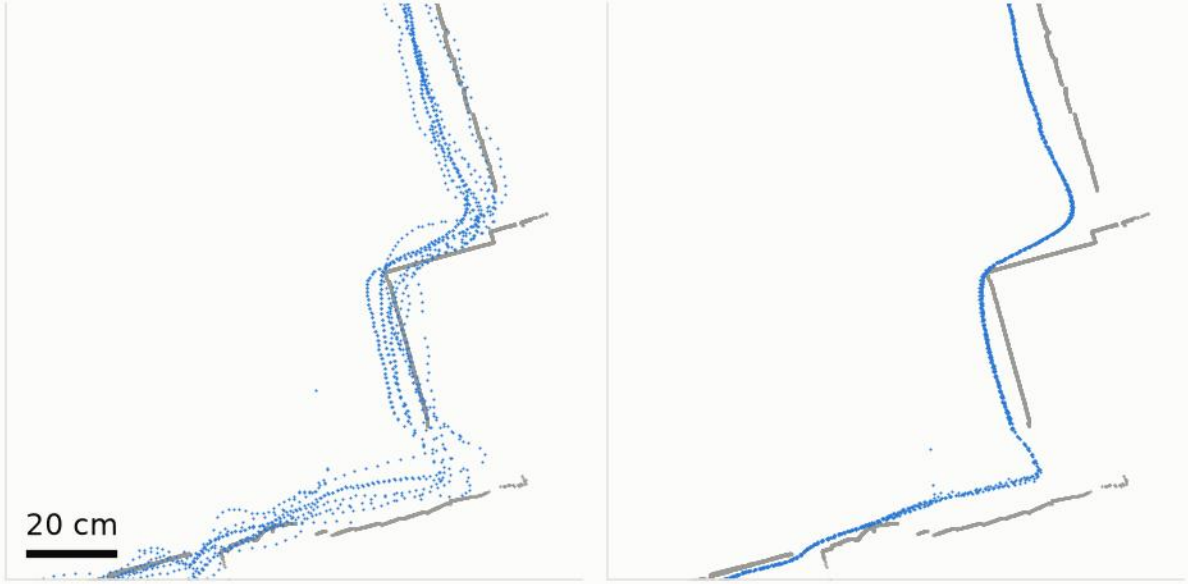
Normals: 6.6° → 4.0° to the scan (DSINE alone 6.8°). Depth: mean error 1.9 % → 2.0 %, no gain. BiNI: 0.3 s.

ScanNet++ 5ee7c22ba0, view 13 (Déjà View K = 16, 20 views); ground truth: ScanNet++ laser-scan mesh rendered into the view. Angles: median over pixels with ground truth, away from its holes.

20 views refined together in 2.5 s: one wall instead of many

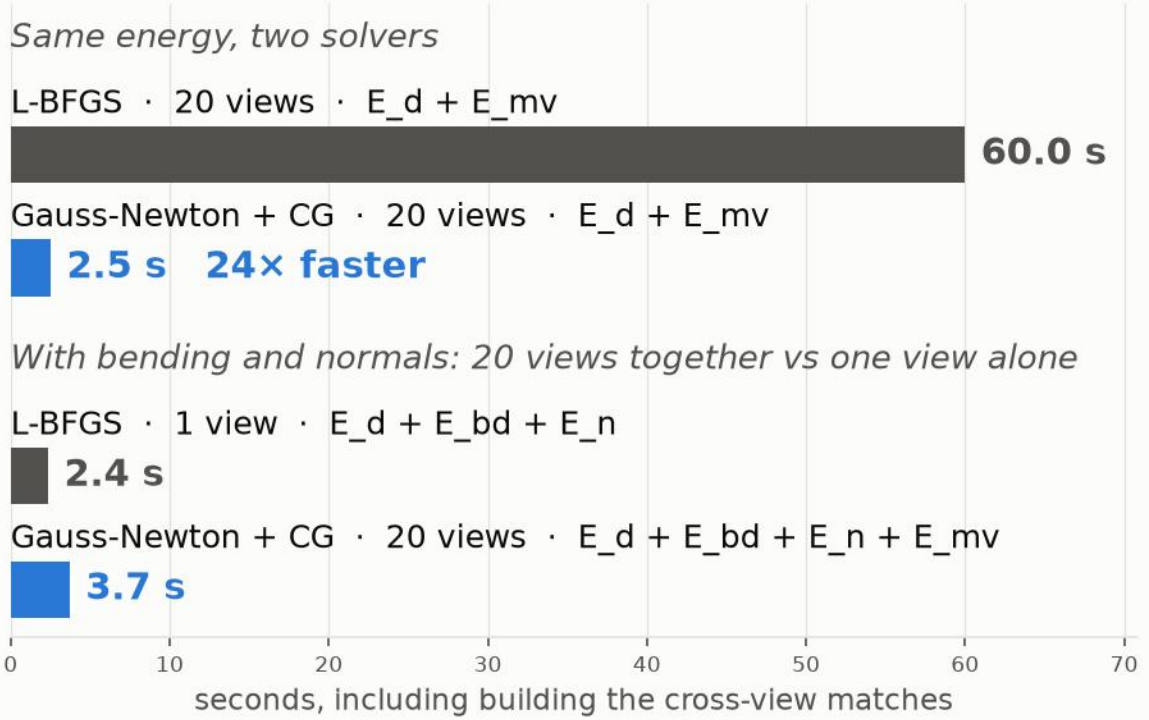
One energy over all 20 depth maps: each stays near Déjà View's prediction, and overlapping views must agree.

before: Déjà View **after: E_d + E_mv**



ETH3D kicker, a 2 cm horizontal slice seen from above:
gray laser scan, blue all 20 depth maps fused

Runtime (RTX 3080, ETH3D office, 336 x 504 px per view)



Energy terms

E_d: stay near Déjà View's depth E_bd: bending (smoothness)
E_n: follow DSINE's normals E_mv: overlapping views agree in depth

E_d + E_mv against Déjà View, ranges over office and kicker:

15-16×
better agreement between views

3-5×
thinner surfaces in the fused cloud

8-10 %
closer to the laser scan, only

Déjà View (K = 16) on 20 views each of ETH3D office and kicker; ground truth: ETH3D laser scans (CC BY-NC-SA 4.0). L-BFGS on 20 views: 500 iterations; Gauss-Newton: 10 steps of 20 CG iterations.